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Received: 24 March 2021 | Accepted: 2 August 2021

DOI: 10.1111/tbed.14277

Transboundary and Emerging Diseases

WILEY

ORIGINAL ARTICLE

Dynamic of *Mycobacterium avium* subspecies *paratuberculosis* infection in a domestic–wildlife interface: Domestic sheep and guanaco as reservoir community

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Funding information
FONDECYT, Grant/Award Number: 1171039; CONICYT, Grant/Award Number: 72140205

Abstract

Natural herbivore populations have experienced uninterrupted pressures from direct and evident domestic-wildlife interactions and competition, to indirect or less obvious ones such as pathogen transmission. Thus, pathogen spillover between wild and domestic animals is a constant concern because the domestic–wildlife interface represents the ecological frontier in which pathogen transmission takes place in both directions. In Patagonian steppe communities, extensive sheep ranching and guanaco (*Lama guanicoe*) populations coexist, and guanaco have shown to be infected by pathogens such as *Mycobacterium avium* subspecies *paratuberculosis* (MAP) likely transmitted from livestock. MAP causes chronic enteritis and affects mostly domestic ruminants. We evaluated MAP prevalence and pathogen shedding in both species' faeces collected in non-shared and shared sites according to presence/absence of sheep and guanaco along a year, in four different seasons (autumn, winter, and spring 2018, and summer 2019). Our results indicate that MAP circulates in both sheep and guanaco populations with self-sustained transmission; however, both species differ in their levels of competence. We detected higher pathogen shedding in sites occupied by sheep, suggesting that sheep populations may be the main source of infection for susceptible animals due to their large numbers which drive MAP dynamics.

KEYWORDS
host competence, infectious disease, pathogen shedding, prevalence

1 | INTRODUCTION

Native herbivores are under important ecological pressures as humans use natural ecosystems (Gordon, 2009). Pressures range from direct and evident domestic-wildlife interactions such as competition for land, to indirect ones such as pathogen transmission (Gordon, 2009). Different host species respond in different ways to the same pathogens, resulting in heterogeneities that affect epidemic dynamics and management (Gervasi et al., 2015; Martin et al., 2019).

Heterogeneities in host–pathogen interactions occur at different scales, from within-organismal levels, to populations and communities. For instance, individuals in a host population may differ in their competence to acquire and transmit an infection to susceptible hosts because of behavioural and physiological traits that influence trans-

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mission (Martin et al., 2019). Some hosts (i.e., individuals, population, species) may be highly competent (Martin et al., 2019), for example, super-spreader hosts at the individual level, or at the species level, by one species acting as reservoir, and therefore source of infection to other species through pathogen spillover (Martin et al., 2019; Plowright et al., 2017). Thus, host competence can be defined as the ability of an infected host to transmit the infection to a susceptible host (Gervasi et al., 2015), as a result. Host competence is a composite trait that depends on duration of infectious period, pathogen shedding, and contacts that can result in infection (Martin et al., 2019). How these different components weigh will vary across host–pathogen systems (e.g., social contacts may not be highly relevant for environmentally transmitted pathogens). Measuring competence in natural populations, as other host traits, is challenging, especially because time of infection is commonly unknown. However, in those cases, random sampling of hosts traits (i.e., pathogen shedding) is feasible and valuable as a "point" proxy of competence at the population level (Martin et al., 2019), in homologous similarity to "point tolerance" to assess tolerance to infection in a population of hosts (Jackson et al., 2014).

Pathogen spillover between wild and domestic animals is a constant concern for ranchers, veterinarians, and wildlife managers (Caron et al., 2013; McClure et al., 2019). The domestic–wildlife interface is the frontier where pathogen transmission takes place in both directions, from domestic reservoirs to wildlife and the reverse, thus affecting the viability of wildlife as well as the economic performance of livestock productions (e.g., Becker et al., 2020; Calenge et al., 2021). When a domestic–wildlife interface occurs, the degree of pathogen transmission between individual hosts and circulation in their populations is eventually determined by behavioural and physiological components at the individual and species level, influencing patterns of infection at the population level, named as "pathogen pressure" (Gervasi et al., 2015). Species phylogenetically close, or that exploit similar ecological niches, have higher probability of sharing pathogens (Plowright et al., 2019). This is especially important in extensive livestock production, where domestic animals and wildlife interact in diffuse spatial borders, facilitating direct or indirect interactions leading to pathogen spillover and subsequent establishment of a multi-host system (McClure et al., 2019). In these newly established systems, pathogen dynamics will be dominated by individual and species level of host's competence, driven eventually to persistence, or to extinction of the infection or the hosts, through dynamics of dilution or amplification effects (Wilber et al., 2020).

In the steppes of southern Patagonia, large natural pastures are exploited for extensive sheep ranching (Oliva et al., 2012). The same geographical area is also used by guanaco (*Lama guanicoe*), a native camelid widely distributed in South America, and the largest wild herbivore on the continent (González et al., 2006). When Europeans first arrived, the guanaco was the dominant herbivore in the southern arid plains and Andes range, in total numbers of ≈30–50 million animals (Raedeke, 1979). The colonization of Patagonia and arrival of livestock species shifted the ecological community to 10 million sheep dominating the landscape (Oliva et al., 2012), while 1 million guanacos use the remnant areas (Baldi et al., 2016). Both species compete directly for food (Pontigo et al., 2020), creating direct conflict between ranchers and guanaco, but also increasing chances for pathogen transmission (Bozzuto et al., 2020). This livestock–guanaco spillover has been proven for several pathogen groups, such as nematodes: *Nematodirus sp.*, *Dictyocaulus filaria*, *Trichuris tenuis;* protozoa: *Eimeria* spp., *Sarcocystis sp.*; cestode: *Moniezia expansa* (Beldomenico et al., 2003); virus: rotavirus (Parreño et al., 2008); and bacteria: *Mycobacterium avium paratuberculosis* (Salgado et al., 2009). The latter is the etiological agent of paratuberculosis, a chronic infection mostly affecting ruminants, and characterized by enteritis (Manning & Collins, 2001).

MAP is a multi-host pathogen, highly resistant to environmental conditions, able to survive in pastures for more than six months (Whittington et al., 2004). The faecal–oral route is the most important MAP transmission mode (Manning & Collins, 2001). After replication in intestinal macrophages, bacteria are shed in faeces; therefore, heterogeneities in faecal shedding at individual, population, and species levels are major determinants of MAP transmission and persistence in a host population or community (Sweeney et al., 2012). Disease signs are often correlated with shedding; therefore, the number of viable pathogens in faeces is a proxy for disease progression and severity (Mitchell et al., 2015). The progression of MAP infection has been mainly characterized in livestock, where it induces clinical signs in old animals with potential economic implications for cattle by decreasing productive performance (García & Shalloo, 2015). MAP impacts seem largely underestimated in wild animals (Carta et al., 2013; Zhu et al., 2019), with incidental reports of health compromise (e.g., *post-mortem* exam in mishmi takin *Budorcas taxicolor*, Naylor et al., 2018) or selective monitoring of diseases (e.g., Iberian ibex *Capra pyrenaica*, Gómez-Guillamón et al., 2020). The chronic progression of MAP infection might be highly relevant in wildlife because it may compromise detection of infections and "hide" its population impacts. Although this pathogen has also been isolated in several wild native species (Corti et al., 2020; Salgado et al., 2015), and the introduced European hare (*Lepus europaeus*; Salgado et al., 2011), no study up to date has assessed the contribution of different domestic and wild species as hosts for the infection and the eco-epidemiological consequences.

Here, we used a longitudinal approach to assess the dynamics of sheep and guanaco as hosts of MAP. Our aims were two-fold: first, we assessed whether MAP prevalence in guanaco and sheep differ according to their spatial and temporal overlap in shared and non-shared areas; and second, we assessed their levels of competence as hosts and the importance of ecological context on their levels of pathogen shedding.

2 | MATERIAL AND METHODS

2.1 | Study sites

The study area included six sites defined as extensive sheep ranching with and without guanaco presence, and national parks (only guanaco), located in Magallanes District, Chilean Patagonia (Figure 1). Two of the sites are located on Tierra del Fuego Island, Ranches 1 (≈2500 ha) and 4

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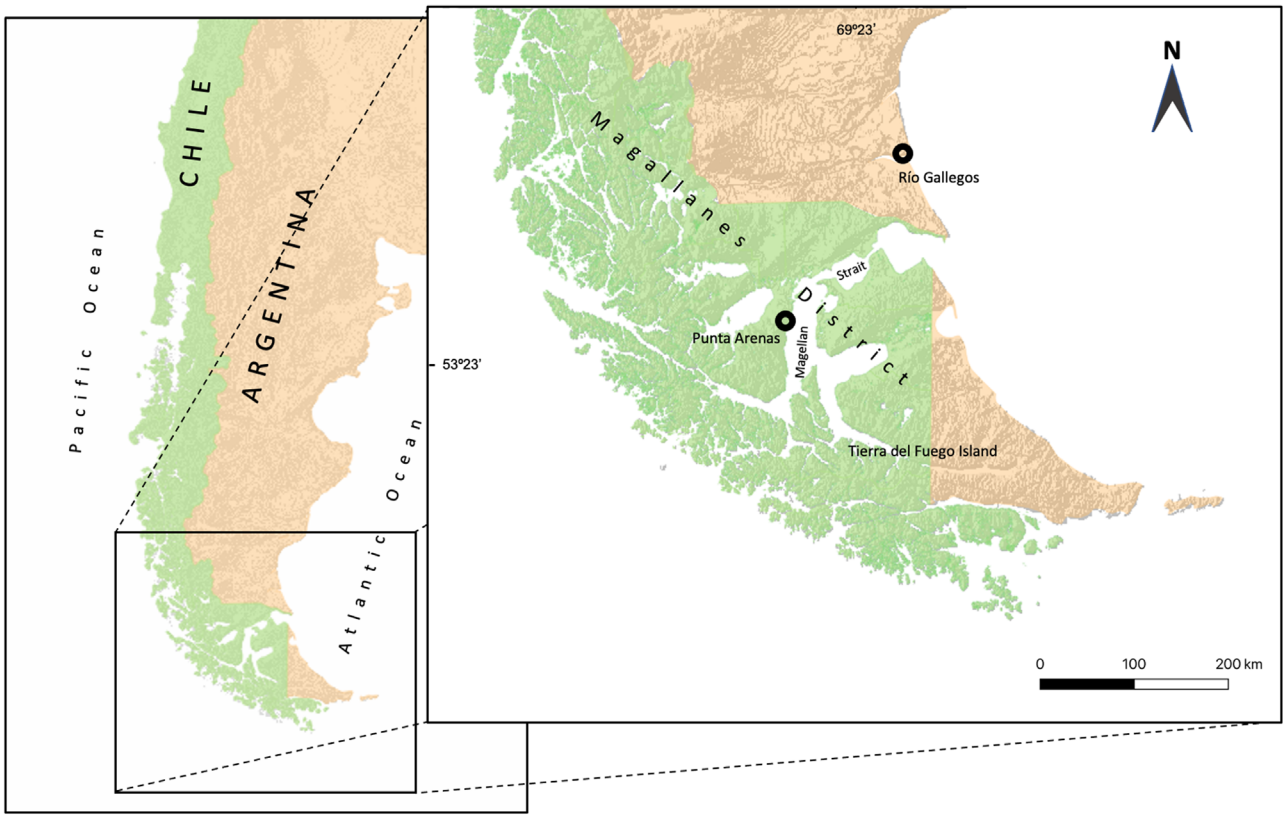


FIGURE 1 Geographical localization of Magallanes district in southern Chilean Patagonia. The six study sites are located in this district

(≈3000 ha), and four sites on the mainland, Ranches 2 (≈2500 ha) and 3 (≈70,000 ha), and Pali Aike (PNPA, ≈5000 ha) and Torres del Paine (PNTP, ≈180,400 ha) national parks. Sheep and guanaco are present in Ranches 3 and 4, representing the "shared" sites treatment. Ranches 1 and 2 have only sheep, while in PNPA and PNTP only guanaco are present, representing the "non-shared" sites.

The climate of the study area is semi-arid and cold, with average temperatures ranging from −4°C in winter to +10°C in summer, and average annual precipitation of 500 mm (Dirección Meteorológica de Chile, 2008). Dominant steppe plant species are herbaceous vegetation such as tussock grass (*Festuca gracillima*) and steppe bushes such as black bush (*Mulguraea tridens*), fachine (*Chiliotrichum diffusum*), green bush (*Lepidophillum cupressiforme*), and diddle-dee (*Empetrum rubrum*). There are also cultivated areas with white clover (*Trifolium repens*) and perennial ryegrass (*Lolium perenne*) (Pontigo et al., 2020).

2.2 | Study design and data collection

We used a longitudinal design to assess the temporal dynamics of MAP prevalence and shedding in two ecological contexts: single-host and mixed-host species. Faecal samples were collected during four sampling periods: Austral autumn (May–June), winter (August–September), spring (November–December; all three seasons in 2018), and summer (January–February, 2019). Samples were collected along three to six walked transects of ≈0.5 to 5 km in length to cover a similar surface per area. A minimum distance between samples >50 m was used to reduce chances of sampling faeces from the same specimen (Gómez-Vinassa & Nuñez, 2016). Pellet size and pile disposition were used as visual cues to differentiate sheep and guanaco faeces. Guanaco faeces are larger than sheep faeces, pellets are elongated, and they defecate in dung-piles (Marino, 2018). Only fresh faeces were collected. Freshness was assessed by the presence of fresh mucus and moisture on the pellets. We prevented environmental contamination at the moment of collection by selecting samples from the top of a pellet pile and/or in no direct contact with grass or the soil, and by using new sterile nitrile gloves for each faecal pile. Samples were stored in hermetic plastic bags and at 5°C until analysed.

At the same time of sample collection, host abundances were estimated through the kilometric abundance index or the animals counted per transect length (number of observed animals·km^{-1}; Ogutu et al., 2008). One to two persons conducted field surveys to count sheep and guanaco within each study area along the same transects used for faecal collection. Each study area was surveyed four times, one time each season. Three to six variable length transects (0.5 to 5 km) per area were surveyed on foot during daylight hours to cover the total of the surveyed area. We estimated host abundance available for MAP, considering sheep and guanaco separately, but also as a community of host herbivores, adding both species together as a measure of total abundance. In the case of the two sites with only sheep, we recorded

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age composition of the herds, which were all females, differentiating among young ewes (<2 years), adult ewes (>2 years), and mixed herds containing both age categories. In the case of guanaco, we identified maternal—adults with young animals (< 2 years)—and adult groups. However, those guanaco groups were present in all areas; therefore, we did not differentiate them as we did with sheep.

2.3 | Bacteriological analysis

Faecal sample analysis followed a three-stage laboratory process to obtain prevalence and pathogen shedding data. Samples were processed and cultured in the BACTEC-MGIT 960 (ParaTB MGIT, Becton Dickinson, Sparks, Maryland, USA) liquid culture system at 37°C for up to 49 days, according to the manufacturer's protocol. This culture method presents a sensitivity level of 73% (CI: 67–78; Cruciani et al., 2004). Positive culture tubes were inverted three times before 500 μL was taken to follow with the DNA extraction process to molecularly confirm MAP identity by a qPCR protocol (Salgado et al., 2014).

The amount of viable infectious pathogen present in the samples was estimated based on the concentration of MAP DNA in the qPCR products (Dzieciol et al., 2010). Differentiation between the main MAP strains, cattle (C-type) and sheep (S-type), was performed through PCR-restriction endonuclease analysis (PCR-REA; Marsh et al., 1999; Park et al., 2018). Then, genotyping of MAP strain isolates was conducted by five highly informative mycobacterial interspersed repetitive units-variable number tandem repeats (MIRU-VNTR) targets (292, X3, 25, 47, and 3; Thibault et al., 2007; Verdugo et al., 2014). PCR amplification of each locus was conducted following amplification conditions previously mentioned (Thibault et al., 2007). The repeat numbers of the PCRs were analysed through electrophoresis on 2% agarose gel. The PCR results and MAP strains were identified in the INMV database (Thibault et al., 2007). For this step, we used only samples that provided DNA of high quality—optimal 260/280 DNA purity ratio and A260/A280 purity index of ≈1.8.

2.4 | Statistical analysis

To estimate disease prevalence in sheep and guanaco at each site and season, we used the number of faecal samples positive and negative to the pathogen culture. We built probability estimates of mean and 95% confidence intervals using the Wilson method implemented in the "*binom*" package for R (Dorai-Raj, 2015). Differences in prevalence among site types and species were compared by the overlap of confidence intervals among sites and seasons (Cumming, 2009).

Positive MAP sheds were Log_{10}-transformed from the equivalence genome to normalize their distribution (Keene, 1995), using as a unit the number of copies of the MAP gen IS*900* (Log_{10} DNA copies/2 g faecal material). Normality was checked through *Q-Q* plots and Shapiro–Wilk tests, outliers were removed to fulfil normality. We used a *t*-test to compare total MAP shedding between sheep and guanacos. MAP shedding by species according to seasons and site types was assessed through three-way ANOVA and Tukey post hoc test within the "*lsr*" package for R (Navarro, 2015).

To assess whether sheep and guanaco might present similar levels and dynamics of competence as MAP hosts, we evaluated the relative importance of predictor variables on MAP shedding for both species. We included biological plausible combinations and interactions among variables in linear models, including single, additive, and pairwise interactions through the function "dredge" in the package "*MuMIn*" for R (Barton, 2019). We formulated independent sets of linear regression models for sheep and guanaco, and the herbivore community (both species together). Model formulations included MAP shedding as response variable and animal abundance (total and by species) at each study site (individuals·km^{-1}), site type (single-species, or shared), sampling season, and sampled species as predictors. For sheep, the model formulations also included three female age categories as predictor (young ewe, adult ewe, and mixed age group). We ranked models based on their support as indicated by the Bayesian information criterion (BIC) (Schwarz, 1978). If more than one model was equally supported (ΔBIC < 2), the regression model coefficients were averaged, obtaining one set of parameter estimates (Turek & Fletcher, 2012). Predictor variables included in the model formulations were tested for multicollinearity through a correlation matrix with the aim to remove those with high correlation values ($R^2 > 0.7$) with ≥ 2 other variables from the analyses (Dormann et al., 2013). No strong correlations were found. All tests were performed with R v.4.0.4 program (R Core Team, 2021), with an $\alpha = 0.05$.

3 | RESULTS

3.1 | Sheep and guanaco population patterns

Host abundance varied across sites and seasons (Table 1). Overall, sheep were more abundant than guanaco, with the exception of one shared and a non-shared site in summer. Sheep abundance in non-shared sites ranged from 5.90 in summer to 285.23 individuals·km^{-1} in autumn, while in shared sites, the abundance ranged from 1.56 in summer to 175.20 individuals·km^{-1} in autumn. For guanaco abundance in non-shared sites, ranges were from 14.43 in spring to 60.67 individuals·km^{-1} in autumn, and in shared sites ranged from, 0.74 in summer to 9.09 individuals·km^{-1} in winter.

Guanaco abundances presented the lowest variability across seasons compared to sheep abundances. The highest guanaco abundance within non-shared sites was about twice the lowest number of animals recorded. In contrast, sheep numbers fluctuated between two- and 24-fold from the lowest number to the highest. These patterns of abundance fluctuations are likely related to the life cycle of the species, driven by natural seasonal variations in guanacos and to the productive cycle and management practices in sheep. The latter is supported with variations of sheep age structure through seasons, presenting mostly young ewes and mixed-age herds in autumn and summer and adult ewe and mixed-age herds in winter and spring.

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TABLE 1 Abundance index (AI) of sheep and guanaco (individuals·km^{-1}) for non-shared and shared sites per sampling season. Pooled AI for shared sites represents sheep plus guanaco numbers per area. Sheep age categories are young ewes (<2 years), adult ewes (>2 years), and mixed age groups formed by both categories

	Autumn 2018		Winter 2018		Spring 2018		Summer 2019	
Site	AI	Group type	AI	Group type	AI	Group type	AI	Group type
Sheep								
Non-shared								
Ranch 1	42.00	Young ewe	121.90	Adult ewe	108.50	Adult ewe	5.90	Young ewe
Ranch 2	285.23	Mixed-age	125.85	Mixed-age	138.14	Mixed-age	225.23	Mixed-age
Shared								
Ranch 3	175.20	Young ewe	44.73	Adult ewe	115.37	Young ewe	144.63	Young ewe
Ranch 4	169.09	Mixed-age	95.22	Adult ewe	83.33	Adult ewe	1.56	Mixed-age
Guanaco								
Non-shared								
Pali Aike	24.19		15.60		14.43		29.91	
Torres del Paine	60.67		26.70		40.95		60.63	
Shared								
Ranch 3	6.20		9.09		7.16		0.74	
Ranch 4	4.82		4.33		1.44		5.33	
Sheep + Guanaco								
	Pooled		Pooled		Pooled		Pooled	
Ranch 3	181.40		53.82		122.53		145.37	
Ranch 4	173.91		99.56		84.78		6.89	

TABLE 2 Seasonal distribution of sheep and guanaco faecal samples collected at six study areas. Data correspond to sample size (*n*), positive samples, and percentage of the total by species and site types (non-shared and shared)

		Seasons								
		Autumn 2018		Winter 2018		Spring 2018		Summer 2019		
Species	Site type	*n*	Positive	*n*	Positive	*n*	Positive	*n*	Positive	Total samples
Sheep	Non-shared	41	2 (4.9%)	40	8 (20.0%)	40	15 (37.5%)	40	2 (5.0%)	161
	Shared	42	14 (33.3%)	39	6 (15.4%)	41	18 (43.9%)	39	31 (79.5%)	161
Guanaco	Non-Shared	48	8 (16.7%)	40	37 (92.5%)	41	15 (36.6%)	40	4 (10.0%)	169
	Shared	40	2 (5.0%)	32	7 (21.9%)	43	15 (34.9%)	38	17 (44.7%)	153
Total per Season		171		151		165		157		644

3.2 | Patterns of MAP prevalence in non-shared and shared sites

A total of 644 faecal samples were collected and cultured for MAP detection (Table 2). Overall, 322 samples corresponded to sheep and 322 corresponded to guanaco. Across all seasons, a total of 201 (64.4%) pathogen cultures were positive to MAP growth (Table 2), with 96 (47.8%) for sheep and 105 (52.2%) for guanaco. MAP prevalence ranged from 5.0% to 92.5%, with variations between host species, seasons, and site type (Figure 2).

Sheep mean prevalence in non-shared sites ranged from 5.0% in summer (CI: 1.4%–16.5%) and 4.9% in autumn (1.4%–16.1%) to 37.5% in spring (24.2%–53.0%). In shared sites, the seasonality of prevalence in sheep presented an opposite dynamic to non-shared sites, being highest in summer with 79.5% (64.5%–89.2%) and lowest in winter with 15.4% (7.3%–29.7%). For guanaco, prevalence in non-shared sites ranged from 10.0% in summer (4.0%–23.1%) to 92.5% in winter (80.1%–97.4%). Winter prevalence was significantly higher than in the other seasons. Intermediate values of prevalence were found in autumn and spring. In shared sites, the seasonality observed contrasted with the prevalence in guanacos in non-shared sites. There was an almost linear increase in prevalence along the year with the lowest value in autumn (mean 5.0%, 1.4%–16.5%) and the highest in summer (44.7%, 30.2%–60.3%).

When comparing the dynamics of both host species in shared and non-shared sites, there seem to be differences in temporality and

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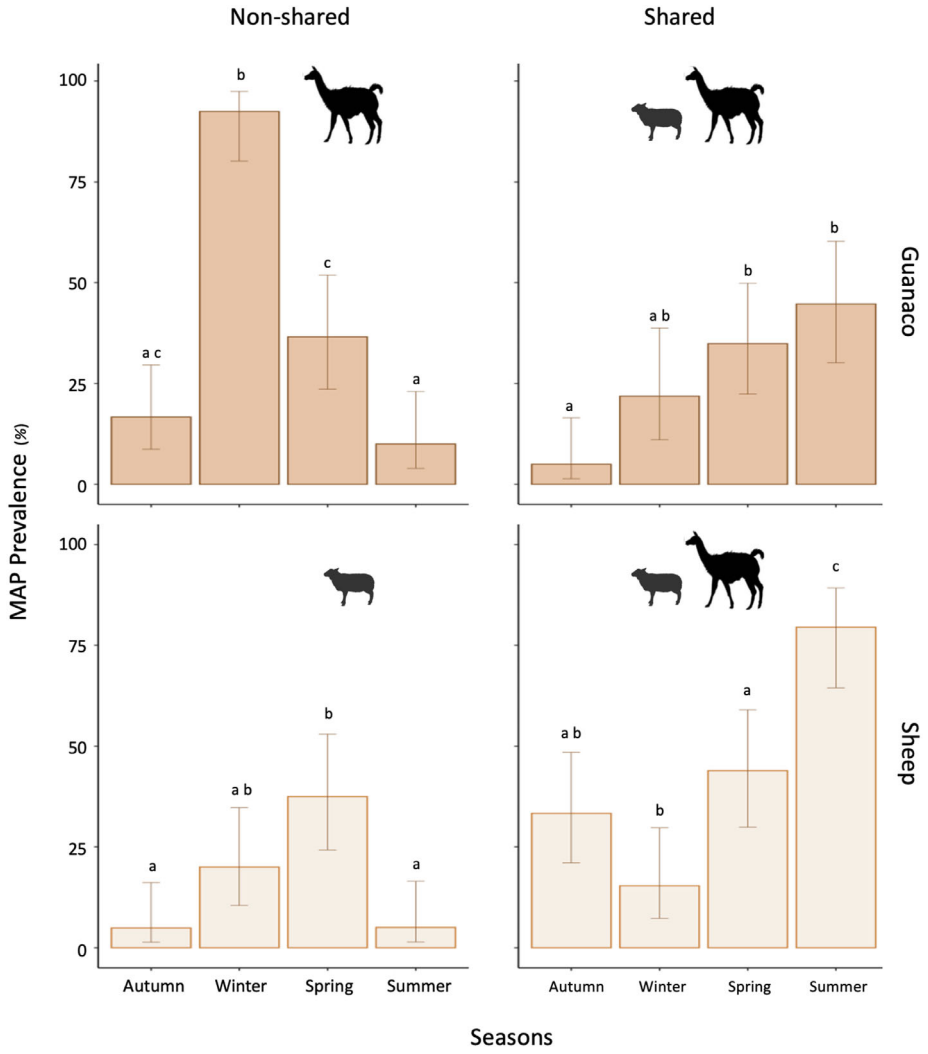


FIGURE 2 Prevalence means and confidence intervals of *Mycobacterium avium* subspecies *paratuberculosis* in sheep and guanaco herds, per season at non-shared and shared sites for sheep and guanaco. Different letters indicate significant differences at each site type per species in different seasons according to 95% confidence interval overlaps

magnitudes of infection rate peaks. For guanaco in non-shared sites, the peak of prevalence in winter seems to be higher than the peak of prevalence in sheep in non-shared sites. In the case of sheep, the highest prevalence was recorded in spring, one season later than in guanaco, but the magnitude was lower and also similar to the prevalence in winter. When observing the prevalence in both species in shared sites, both species presented the peak in summer, but the magnitude was much larger in sheep. The seasonal shift in this case seems to be in the moment of the lowest prevalence, which happened in autumn for guanaco, while in winter for sheep, though the confidence intervals overlap.

3.3 | Patterns of pathogen shedding in non-shared and shared sites

Overall pathogen shedding was significantly higher in sheep (mean ± SD: 3.76 ± 0.86 Log_{10} DNA copies/2 g faecal material) than in guanaco (3.26 ± 0.65 Log_{10} DNA copies/2 g faecal material; $t = -4.69$, $df = 199$, $p < .001$). Patterns of seasonal shedding in shared areas were similar to prevalence (Figure 3). Seasonal significant differences were evident in both species (season: $F = 73.15$, $df = 3$, $p < .001$; species: $F = 59.26$, $df = 1$, $p < .001$), including a relationship between season and species present ($F = 9.13$, $df = 3$, $p < .001$), but without relation between site and season ($F = 1.40$, $df = 6$, $p = .216$).

Pathogen shedding in sheep showed a seasonal pattern, being lowest in autumn and higher in summer. In non-shared sites, shedding ranged from 2.33 (Log_{10} DNA copies/2 g faecal material) in autumn to 5.53 in summer. In shared sites, shedding ranged from 2.46 in autumn to 5.46 in summer. Pathogen shedding in guanaco presented a different pattern compared to sheep, being lowest in autumn and spring, and higher in summer. In non-shared sites, pathogen shedding of guanaco ranged from 2.33 in autumn to 4.41 in summer. In shared sites, it ranged from 2.48 in spring and 4.55 in summer.

When comparing the dynamics of both host species in non-shared and shared sites, there was a similar tendency in pathogen shedding

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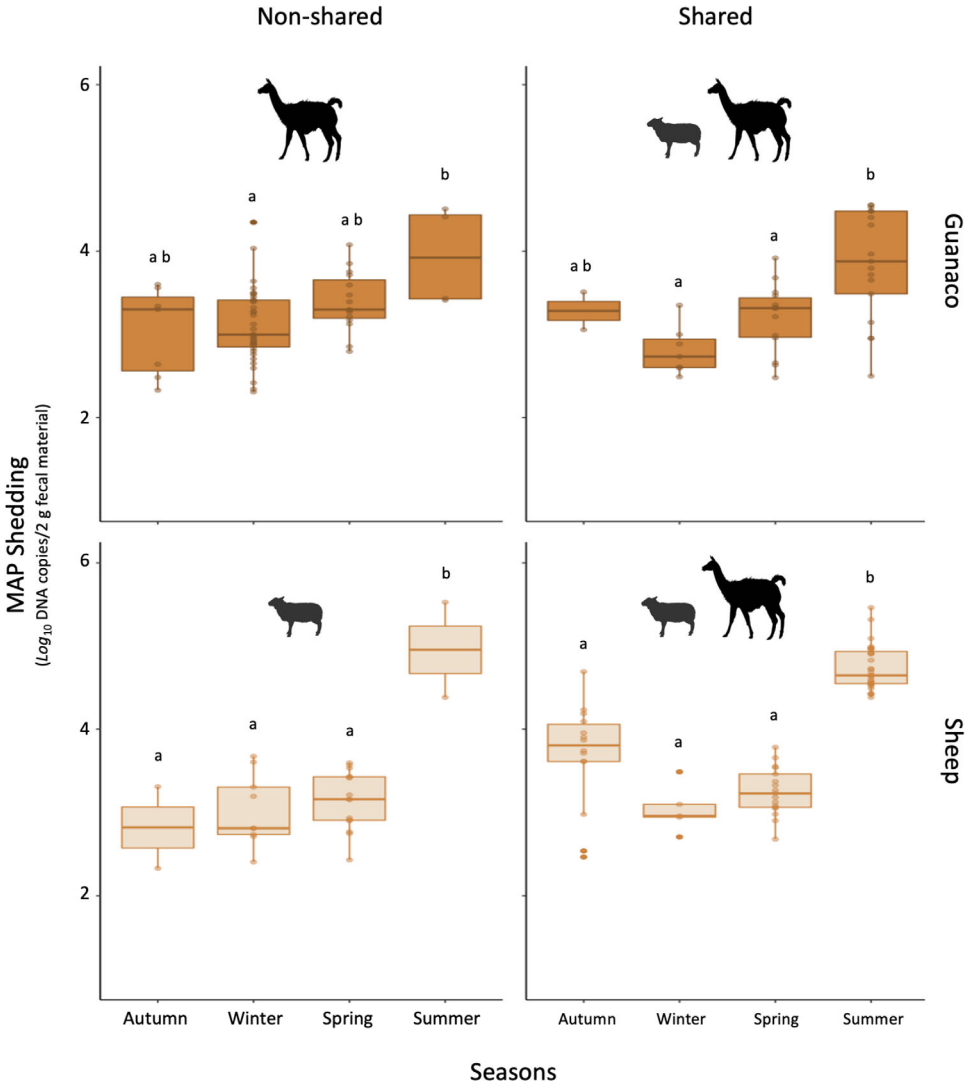


FIGURE 3 *Mycobacterium avium* subspecies *paratuberculosis* shedding distributed by season at non-shared and shared sites for sheep and guanaco. Different letters indicate significant differences at each site by species and seasons

through seasons, linearly growing from autumn to summer. In shared sites, both species present a similar dynamic pattern, with the peak in summer, but the lowest value changed from spring to summer. Pathogen shedding in all seasons and sites were always larger in sheep than in guanaco.

3.4 | Variable shaping host competence for MAP in the herbivore community

When analysing pathogen shedding in the herbivore host community as a whole—sheep and guanaco together—to assess whether both species present similar levels of host competence for MAP, only one model formulation was the most supported. The retention of host species in the most supported model suggested that sheep and guanaco differ in their levels of competence as hosts; specifically, sheep shed higher numbers of pathogens into the environment than do guanaco.

In the second step, to analyse similarities in the relevance of ecological variables to the shedding dynamics of both host species separately, our model selection approach showed that the importance of explanatory variables varied according to the host and their ecological context (Table 3). The model coefficient estimates showed that although abundance was part of the most supported model, it presented a small negative effect.

Thus, increasing the number of sheep tended to decrease the amount of pathogen shed at the individual level. Age structure of sheep herds influenced MAP shedding, with the highest number of pathogens in groups dominated by young animals, followed by mixed-age groups in comparison to ewes. For seasons, winter and spring showed a negative effect on shedding, while the number of pathogens shed increased in summer and autumn.

For guanaco, two additive model formulations were supported, one considered total abundance and season, and the other included sheep abundance and season, as predictor variables. Average model coeffi-

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TABLE 3 Linear models for *Mycobacterium avium* subspecies paratuberculosis shedding in sheep and guanaco. Information corresponds to BIC and its differences (ΔBIC), BIC weight (ω), R^2 value, and number of variables in the model (k). The abundance corresponds to the abundance index (individuals·km^{-1}). Site is non-shared and shared areas used by sheep and guanaco. Season corresponds to autumn, winter, and spring 2018, and summer 2019. The group variable corresponds to young ewe, adult ewe, and mixed age groups

Sheep model	**BIC**		**BICω**	**R^2**	**k**
Total abundance + Season + Group	101.8		0.48	0.82	3
Variables	**Estimates**			**Confidence intervals (95%)**	
(Intercept)	3.62			3.26 – 3.98	
Total abundance	−0.01			−0.00 – −0.00	
Group (young ewe)	0.53			0.26 – 0.80	
Group (mixed age)	0.10			−0.13 – 0.33	
Season (winter)	−0.34			−0.67 – −0.01	
Season (spring)	−0.31			−0.57 – −0.05	
Season (summer)	1.02			0.74 – 1.30	
Guanaco models	**BIC**	**ΔBIC**	**BICω**	**R^2**	**k**
Total abundance + Season	153.4	0.00	0.37	0.32	2
Sheep abundance + Season	153.7	0.28	0.32	0.32	2
Variables	**Estimates**			**Confidence intervals (95%)**	
(Intercept)	3.24			2.89 – 3.59	
Total abundance	−0.00			−0.01 – −0.00	
Season (winter)	−0.12			−0.47 – 0.23	
Season (spring)	0.21			−0.15 – 0.56	
Season (summer)	0.76			0.38 – 1.30	
Sheep abundance	−0.00			−0.00 – −0.00	
Herbivore community models	**BIC**		**BICω**	**R^2**	**k**
Season + Site + Species	293.2		0.65	0.61	3
Variables	**Estimates**			**Confidence intervals (95%)**	
(Intercept)	3.28			3.06 – 3.50	
Season (winter)	−0.21			−0.34 – 0.03	
Season (spring)	−0.03			−0.25 – 0.20	
Season (summer)	1.06			0.84 – 1.29	
Site (sheep)	−0.60			−0.89 – −0.32	
Site (shared)	−0.28			−0.48 – −0.07	
Species (sheep)	0.54			0.35 – 0.73	

cient estimates showed that total guanaco and sheep numbers had a small but negative effect. Thus, increasing either the number of sheep or guanaco tended to decrease the number of pathogens shed. For seasons, winter had a negative effect on shedding, while the number of pathogens shed increased in spring and summer.

3.5 | MAP types

From the total 201 bacterial cultures positive for MAP, 45 met the DNA quality criteria for typification; of these, 32 were successfully typed (Table 4). Overall, MAP C-type (cattle strain) was the most frequently detected across all the typified samples (78%) followed by the S-type (sheep strain; 22%). C-type was detected in both host species regardless of the site, 15 samples (47%) from guanaco, 8 in non-shared and 7 in shared sites, and 10 (31%) in sheep, 5 in non-shared and 5 in shared sites. Three cultures from sheep samples (9%) and one from guanaco (3%) from shared sites were typified as S-type. The S-type was also identified in three guanaco samples from non-shared sites (9%). Genotyping using the MIRU-VNTR method differentiated 10 subtypes among the 32 MAP isolates (Table 4). The most prevalent type was INMV 2 (62.5% isolates), then INMV 77 (9.4%) and INMV 24 (6.3%). INMV 13, 74, New-1, New-2, New 3, New-4, and New-5 represent 3.1% of isolates of each one. Eighteen C-type strains were identified as INMV 2 and two as S-type. Two C-type and one S-type were INMV 77.

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TABLE 4 MIRU-VNTR profiles of 13 sheep and 19 guanaco MAP isolates from typed faecal samples. Genotyping differentiated 10 INMV subtypes, distributed in C-type (cattle strains) and S-type (sheep strain)

	Hosts (*n* isolates)		Site type		Number of MIRU-VNTR copies						
INMV	Sheep	Guanaco	Non-shared	Shared	292	X3	25	47	3	Numerical Code	Strain Type
INMV 2	10	10	10	10	3	2	3	3	2	32332	18C 2S
INMV 13	–	1	1	–	2	2	3	3	2	22332	C
INMV 24	–	2	–	2	3	2	1	3	2	32132	1C 1S
INMV 74	–	1	–	1	3	3	3	3	2	33332	C
INMV 77	–	3	3	–	3	2	2	3	2	32232	2C 1S
INMV-New 1	–	1	–	1	3	3	5	3	2	33532	C
INMV-New 2	–	1	–	1	3	2	3	11	9	323119	C
INMV-New 3	1	–	1	–	3	1	2	3	2	31232	C
INMV-New 4	1	–	1	–	1	3	3	3	2	13332	S
INMV-New 5	1	–	–	1	3	3	5	3	2	33532	S

4 | DISCUSSION

We present the first longitudinal study of the infection dynamics of *Mycobacterium avium* subspecies *paratuberculosis* in a sympatric domestic and wild herbivore interface in the Patagonian steppe. Our results indicate MAP circulation in both sheep and guanaco populations with similar infection rates; however, the seasonal patterns seem to differ accordingly to the presence of co-grazing or not. MAP was widespread in all and across study sites as observed in other geographical locations where this infection is present (e.g., Eisenberg et al., 2010; Verdugo et al., 2014), not showing a clustered distribution as observed in some wild species (i.e., Judge et al., 2020). The fact that the overall percentage of positive MAP samples was similar between sheep and guanaco suggests that MAP might have found a way to perpetuate itself within this mixed herbivore community in Patagonia.

Seasonal patterns of prevalence in sheep and guanaco seemed to depend on their ecological context. Differences in temporality and magnitude of peak prevalence between sheep and guanaco in non-shared sites support potential distinction in transmission dynamics (Rhyan et al., 2013). In guanaco, the highest prevalence was detected in winter, while in sheep, the peak prevalence was in spring. It is possible that this pattern may reflect the life cycle of both species, affecting their behaviour and physiology. Sheep present seasonal movements between winter and summer grazing ranges with higher animal concentrations in spring (Pontigo et al., 2020), coinciding with the lambing period. In contrast, guanaco aggregate in higher densities in winter, before splitting into maternal and bachelor groups in summer (González et al., 2006). Those stressing periods may promote MAP shedding in animal faeces, changing prevalence patterns (Beate et al., 2005). It is relevant to mention that the prevalence of MAP in guanaco reported here is among the highest reported in non-domestic species (Carta et al., 2013). In the Iberian ibex (*Capra pyrenaica*), mean MAP prevalence is 1.87 (CI: 0.05–3.68) (Astorga-Márquez et al., 2014) and in the Cantabrian chamois (*Rupicapra pyrenaica*), mean prevalence is 9.7%, (0.0–13.5) (Falconi et al., 2010). However, diagnostic sensitivity of the liquid culture method is not 100% (Cruciani et al., 2004), then prevalence is perhaps underestimated, and infection rates could be higher among both ungulate species (Whittington & Sergeant, 2001).

These seasonal trends changed when sheep and guanaco co-occurred in an area, when both species presented the highest prevalence in summer. Interestingly, the peak prevalence in sheep was higher in shared sites, while the peak in guanacos in these sites was lower than in single species sites. Thus, the presence of the other species in the host community affects the probability of infection. One potential hypothesis explaining this pattern is behavioural and ecological changes between sheep and guanaco, which may result in occupancy of different areas within the sites. It has been argued that guanaco spatially segregate from sheep in shared sites (Iranzo et al., 2013), and also that fewer individual guanacos use shared areas and even reduce their trophic niche in such areas (Pontigo et al., 2020). In the case of sheep presenting higher prevalence in shared areas, we hypothesise that this pattern may reflect the age composition of the groups in those areas, which was dominated by young animals, compared to sheep-only sites in which the groups presented mixed-age composition. Peaks of prevalence in groups of young sheep in the summer reflect the influx of young susceptible individuals transitioning to a grass-based diet, with higher risk of infection. This hypothesis is supported by the higher number of viable pathogens shed in summer. In the case of guanaco, the higher prevalence related to their longer lifespan than sheep (<6 years; Texeira & Paruelo, 2006), reaching a maximum of 30 years (Wheeler, 1991), allowing for the typical chronic progression of this disease (Mitchell et al., 2015; Whittington & Sergeant, 2001).

The genotyping of positive samples showed a high diversity of genotypes supporting different clonal origins of the strains circulating in the study areas. Not surprisingly, most of the samples genotyped were

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classified as INVM-2, which is one of the most worldwide distributed genotypes (i.e., Biet et al., 2012; Douarre et al., 2011; Imperiale et al., 2017; Stevenson et al., 2009). Of the other four genotypes recorded in the reference database, two have been reported (INVM-13 in Europe, Stevenson et al., 2009; Thibault et al., 2007; Argentina, Gioffré et al., 2015; INVM-24 in Europe, Stevenson et al., 2009), the remnant two did not include geographical information. Although in low frequency, we report five new genotypes classified as C/S types, which were present in single samples from guanaco (INVM-New 1 and 2) and sheep (INVM-New 3, 4, and 5). The detection of these genotypes supports potential differentiation of clonal lineages in the herbivore community of Patagonia, but further studies, using molecular methods with higher resolution, would prove valuable to assess potential adaptation of bacteria to guanaco and the directionality of transmission. Although C-type MAP seems the dominant strain in the area (Salgado et al., 2009), suggesting cattle as the primary source of MAP infection, we cannot rule out sheep as another source of infection, since C-type can also infect this species (Kennedy & Benedictus, 2001; Muskens et al., 2001).

The intensity of pathogen shedding was higher in both species in summer. Overall, shedding was higher in sheep than in guanaco across all seasons, supporting a higher competence as hosts for MAP with a potentially higher role in driving pathogen pressure in the community (Gervasi et al., 2015; Plowright et al., 2017). Considering that shedding is associated with infection severity in hosts (Dukkipati et al., 2016), the potential for clinical disease was expected to be higher in sheep than in guanaco; however, no signs of clinical disease were detected in either host species. The higher shedding at the individual level in sheep and their higher abundance in relation to guanaco, support sheep as the most relevant host for MAP in the sheep-guanaco system, despite MAP infection perhaps being self-sustained in the guanaco population (Heuer et al., 2012). Although seasonal variation of MAP shedding may relate with animals' physiological stress affecting, for instance, their immune system, such as winter food shortage and harsh conditions in wildlife (Strandin et al., 2018), or reproductive pressure in domestic livestock (Laurin et al., 2015; Strickland et al., 2005), MAP infection often persists in ruminant hosts from 2 to 5 years before onset of clinical disease (Whittington et al., 2004), efficiently invading, replicating, and laying siege to the host without being detected or cleared by the immune system (Harris & Barletta, 2001; Strickland et al., 2005). In addition, climate conditions in southern Patagonia are not as harsh as in the northern hemisphere at the similar latitude, with mild winters and cool summers, where environmental conditions for MAP survival are optimal (Whittington et al., 2004).

MAP infection progression is chronic in ruminants, starting with early transmission into neonates and young animals; however, shedding starts as adults and continues throughout their lifetime (Manning & Collins, 2001). MAP dissemination differs between adult and young animals, with constant and higher shedding in adults, whereas in the young, it is intermittent as the infection progresses (Manning & Collins, 2001). In contrast to the age-dependent pattern of shedding reported in other studies, our model selection approach for sheep evidenced that shedding was higher in younger flocks. This could result from at least two potential scenarios: first, that young individuals might present higher shedding due to early infection from highly contaminated pastures (Mitchell et al., 2015), and second, young animals may start grazing earlier than in other regions due to low quality pastures, facilitating infection, and shedding at younger ages (Ezenwa, 2004). On the one hand, the first scenario may result from the management of the herds in Patagonia, in which weaned female lambs stay for more time in the birthing pastures, while the ewes are moved to summer ranges (Texeira & Paruelo, 2006). On the other hand, the second scenario is supported by evidence of overgrazing of Patagonian pastures by livestock (Gaitán et al., 2018), with a shorter lactation period (Texeira & Paruelo, 2006), resulting in high mortality of lambs in some cases due to poor condition (Hall & Paruelo, 2006). In both cases, managing animal density could modify the epidemic dynamic of MAP.

Guanaco was once the most abundant large herbivore of South America; however, its populations have been reduced, with several local populations becoming extinct (Vilá & Arzamendia, 2020). Whether direct conflict and competition with livestock have been important in guanaco population declines remain unclear, but here we added new evidence of pathogens spillover from livestock to guanaco that might have contributed to population reduction, as it has occurred in other large wild herbivores (Heard et al., 2013; Pedersen et al., 2007). Currently, Patagonian guanaco represents the largest continuous population of this camelid (González et al., 2013), and contact with large numbers of livestock seems a more recent phenomenon than in the rest of this species' range (Baldi et al., 2004). Consequently, some impacts may not be visible at individual or population levels. It is uncertain if infectious diseases negatively affect guanaco populations, as had happened with brucellosis in bison (*Bison bison*) and elk in North America (Rhyan et al., 2013), or if guanaco have become a reservoir for some diseases. Pathogen transmission between wild and domestic herbivores in extensive management conditions, as in the sheep-guanaco interface, could result in the conformation of a new host community for infections like MAP (Nthiwa et al., 2019).

5 | CONCLUSIONS

Our study presents the first comparative assessment of competence of guanaco and sheep as hosts for MAP, demonstrating the circulation of the infection in both species, and also that their dynamics change when considered as a host community. Further studies with increased replication of sites and ecological conditions will be valuable to understand the dynamics of infection. We present our results as early indicators of potential trends in the host community, though we recognize that low sample sizes and replication may affect the generalization of our conclusions. It would also be necessary to re-evaluate, in future research, aspects related to MAP infection pathogenicity in sheep and guanaco, and MAP evolutive adaptation to both hosts, which have different life-history and population dynamic patterns.

ACKNOWLEDGEMENTS

This work was funded by FONDECYT 1171039 granted to P. Corti, S. Radic, and M. Salgado. C.A. Moraga was funded by CONICYT

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BecasChile 72140205. We thank Chile's National Forestry Service (CONAF) for supporting fieldwork in their administrative areas. Special thanks to the parkwardens of Pali Aike and Torres del Paine national parks. We also thank the ranch owners that granted us access to their properties. Special thanks to the IDEAL research center for the use of their facilities in Punta Arenas. We thank field assistants D. Santander, L. Bahamonde, F. González, D. Sepúlveda, T. Fernández, A. Lancaster, K. Paulsen, F. Saiter, C. González, and D. Opazo. C. Riquelme, A. von Hardenberg, and S. Estay for statistical suggestions.

ETHICS STATEMENT

The authors confirm that the ethical policies of the journal, as noted on the journal's author guidelines page, have been adhered to and the appropriate ethical review committee approval has been received from the Bioethical Committee of the University Austral of Chile, Reference Number: 282/2017.

CONFLICT OF INTEREST

The authors declare no conflict of interests.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

Astorga-Márquez, R. J., Carvajal, A., Maldonado, A., Gordon, S. V., Salas, R., Gómez-Guillamón, F., Sánchez-Baro, A., López-Sebastián, A., & Santiago-Moreno, J. (2014). Influence of cohabitation between domestic goat (*Capra aegagrus hircus*) and Iberian ibex (*Capra pyrenaica hispanica*) on seroprevalence of infectious diseases. *European Journal of Wildlife Research*, *60*, 387–390. https://doi.org/10.1007/s10344-013-0785-9.

Baldi, R., Acebes, P., Cuéllar, E., Funes, M. C., Hoces, D., Puig, S., & Franklin, W. L. (2016). The IUCN Red List of Threatened Species 2016. *Lama guanicoe*. [Online] https://doi.org/10.2305/IUCN.UK.2016-1.RLTS.T11186A18540211

Baldi, R., Pelliza-Sbriller, A., Elston, D., & Albon, S. (2004). High potential for competition between guanacos and sheep in Patagonia. *Journal of Wildlife Management*, *68*, 924–938. https://doi.org/10.2193/0022-541X(2004)068[0924:HPFCBG]2.0.CO;2.

Barton, K. (2019). Package 'MuMIn' version 1.9.18. http://mumin.r-forge.r-project.org/MuMIn-manual.pdf

Becker, D. J., Ketterson, E. D., & Hall, R. J. (2020). Reactivation of latent infections with migration shapes population-level disease dynamics. *Proceedings of the Royal Society B*, *287*, 20201829. https://doi.org/10.1098/rspb.2020.1829.

Beldomenico, P. M., Uhart, M., Bono, M. F., Marull, C., Baldi, R., & Peralta, J. L. (2003). Internal parasites of free-ranging guanacos from Patagonia. *Veterinary Parasitology*, *118*, 71–77. https://doi.org/10.1016/j.vetpar.2003.09.008.

Bozzuto, C., Schmidt, B. R., & Canessa, S. (2020). Active responses to outbreaks of infectious wildlife diseases: Objectives, strategies, and constraints determine feasibility and success. *Proceedings of the Royal Society B*, *287*, 20202475. https://doi.org/10.1098/rspb.2020.2475.

Biet, F., Sevilla, I. A., Cochard, T., Lefrançois, L. H., Garrido, J. M., Heron, I., Juste, R. A., McLuckie, J., Thibault, V. C., & Supply, P. (2012). Inter- and intra-subtype genotypic differences that differentiate *Mycobacterium avium* subspecies *paratuberculosis* strains. *BMC Microbiology*, *12*, 264. https://doi.org/10.1186/1471-2180-12-264.

Calenge, C., Lambert, S., Petit, E., Thébault, A., Gilot-Fromont, E., Toïgo, C., & Rossi, S. (2021). Estimating disease prevalence and temporal dynamics using biased capture serological data in a wildlife reservoir: The example of brucellosis in Alpine ibex (*Capra ibex*). *Preventive Veterinary Medicine*, *187*, 105239. https://doi.org/10.1016/j.prevetmed.2020.105239.

Caron, A., Miguel, E., Gomo, C., Makaya, P., Pfukenyi, D. M., Foggin, C., Hove, T., & de Garine-Wichatitsky, M. (2013). Relationship between burden of infection in ungulate populations and wildlife/livestock interfaces. *Epidemiology & Infection*, *141*, 1522–1535. https://doi.org/10.1017/S0950268813000204.

Carta, T., Álvarez, J., Pérez de la Lastra, J. M., & Gortázar, C. (2013). Wildlife and paratuberculosis: A review. *Research in Veterinary Science*, *94*, 191–197. https://doi.org/10.1016/j.rvsc.2012.11.002.

Corti, P., Collado, B., Riquelme, C., Tomckowiack, C., & Salgado, M. (2020). *Mycobacterium avium* subsp. *paratuberculosis* (MAP) infection in the endangered huemul deer (*Hippocamelus bisulcus*) in Patagonia. *Austral Journal of Veterinary Sciences*, *52*, 33–35. https://doi.org/10.4067/S0719-81322020000100106.

Cruciani, M., Scarparo, C., Malena, M., Bosco, O., Serpelloni, G., & Mengoli, C. (2004). Meta-analysis of BACTEC MGIT 960 and BACTEC 460 TB, with or without solid media, for detection of *Mycobacteria*. *Journal of Clinical Microbiology*, *42*, 2321–2325. https://doi.org/10.1128/JCM.42.5.2321-2325.2004.

Cumming, G. (2009). Inference by eye: Reading the overlap of independent confidence intervals. *Statistics in Medicine*, *28*, 205–220. https://doi.org/10.1002/sim.3471.

Dirección Meteorológica de Chile. (2008). Climatology of Chile. Department of Climatology, Subdirection of Climatology and Applied Methodology. Meteorological Directorate of Chile. https://climatologia.meteochile.gob.cl/application/publicaciones/anuario/2007

Dorai-Raj, S. (2015). Package 'Binom'. https://cran.r-project.org/web/packages/binom/binom.pdf

Dormann, C. F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., Marquéz, J. R. G., Gruber, B., Lafourcade, B., Leitão, P. J., Münkemüller, T., McClean, C., Osborne, P. E., Reineking, B., Schröder, B., Skidmore, A. K., Zurell, D., & Lautenbach, S. (2013). Collinearity: A review of methods to deal with it and a simulation study evaluating their performance. *Ecography*, *36*, 27–46. https://doi.org/10.1111/j.1600-0587.2012.07348.x.

Douarre, P. E., Cashman, W., Buckley, J., Coffey, A., & O'Mahony, J. (2011). Molecular characterization of *Mycobacterium avium* subsp. *paratuberculosis* using multi-locus short sequence repeat (MLSSR) and mycobacterial interspersed repetitive units-variable number tandem repeat (MIRU-VNTR) typing methods. *Veterinary Microbiology*, *149*, 482–487. https://doi.org/10.1016/j.vetmic.2010.12.001.

Dukkipati, V. S. R., Ridler, A. L., Thompson, K. G., Buddle, B. M., Hedgespeth, B. A., Price-Carter, M., Begg, D. J., Whittington, R. J., Gicquel, B., & Murray, A. (2016). Experimental infection of New Zealand Merino sheep with a suspension of *Mycobacterium avium* subspecies *paratuberculosis* (MAP) strain Telford: Kinetics of the immune response, histopathology and MAP culture. *Veterinary Microbiology*, *195*, 136–143. https://doi.org/10.1016/j.vetmic.2016.09.018.

Dzieciol, M., Volgger, P., Khol, J., Baumgartner, W., Wagner, M., & Hein, I. (2010). A novel real-time PCR assay for specific detection and quantification of *Mycobacterium avium* subsp. *paratuberculosis* in milk with the inherent possibility of differentiation between viable and dead cells. *BMC Research Notes*, *3*, 251. https://doi.org/10.1186/1756-0500-3-251.

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Eisenberg, S. W. F., Nielen, M., Santema, W., Houwers, D. J., Heederik, D., & Koets, A. P. (2010). Detection of spatial and temporal spread of *Mycobacterium avium* subsp. *paratuberculosis* in the environment of a cattle farm through bio-aerosols. *Veterinary Microbiology*, *143*, 284–292. https://doi.org/10.1016/j.vetmic.2009.11.033.

Ezenwa, V. O. (2004). Interactions among host diet, nutritional status and gastrointestinal parasite infection in wild bovids. *International Journal for Parasitology*, *34*, 535–542. https://doi.org/10.1016/j.ijpara.2003.11.012.

Falconi, C., Oleaga, Á., López-Olvera, J. R., Casais, R., Prieto, M., & Gortázar, C. (2010). Prevalence of antibodies against selected agents shared between Cantabrian chamois (*Rupicapra pyrenaica parva*) and domestic goats. *European Journal of Wildlife Research*, *56*, 319–325. https://doi.org/10.1007/s10344-009-0322-z.

Gaitán, J. J., Bran, D. E., Oliva, G. E., Aguiar, M. R., Buono, G. G., Ferrante, D., Nakamatsu, V., Ciari, G., Salomone, J. M., Massara, V., Martínez, G. G., & Maestre, F. T. (2018). Aridity and overgrazing have convergent effects on ecosystem structure and functioning in Patagonian rangelands: Aridity and grazing effects on rangelands. *Land Degradation & Development*, *29*, 210–218. https://doi.org/10.1002/ldr.2694.

García, A. B., & Shalloo, L. (2015). The economic impact and control of paratuberculosis in cattle. *Journal of Dairy Science*, *98*, 5019–5039. https://doi.org/10.3168/jds.2014-9241.

Gervasi, S. S., Civitello, D. J., Kilvitis, H. J., & Martin, L. B. (2015). The context of host competence: A role for plasticity in host–parasite dynamics. *Trends in Parasitology*, *31*, 419–425. https://doi.org/10.1016/j.pt.2015.05.002.

Giannitti, F., Fraga, M., Caffarena, R. D., Schild, C. O., Banchero, G., Armién, A. G., Travería, G., Marthaler, D., Wells, S. J., & Riet-Correa, F. (2018). *Mycobacterium paratuberculosis* sheep type strain in Uruguay: Evidence for a wider geographic distribution in South America. *Journal of Infections in Developing Countries*, *12*, 190–195. https://doi.org/10.3855/jidc.9751.

Gioffré, A., Correa-Muñoz, M., Alvarado, M. F., Pinedo, R., Vaca, C., Morsella, M., Fiorentino, A., Paolicchi, F., Ruybal, P., Zumárraga, M., Travería, G. E., & Romano, M. I. (2015). Molecular typing of Argentinian *Mycobacterium avium* subsp. *paratuberculosis* isolates by multiple-locus variable number-tandem repeat analysis. *Brazilian Journal of Microbiology*, *46*, 557–564. https://doi.org/10.1590/S1517-838246220140283.

Gómez-Guillamón, F., Díaz-Cao, J. M., Camacho-Sillero, L., Cano-Terriza, D., Alcaide, E. M., Cabezón, O., Arenas, A., & García-Bocanegra, I. (2020). Spatiotemporal monitoring of selected pathogens in Iberian ibex (*Capra pyrenaica*). *Transboundary Emerging Diseases*, *67*, 2259–2265. https://doi.org/10.1111/tbed.13576.

Gómez-Vinassa, M. L., & Nuñez, M. B. (2016). Dieta estacional de guanacos (*Lama guanicoe*) y burros ferales (*Equus asinus*) en un ambiente semiárido de San Luis, Argentina. *Ecología Austral*, *26*, 178–188. https://doi.org/10.25260/EA.16.26.2.0.181.

González, B. A., Palma, R. E., Zapata, B., & Marín, J. C. (2006). Taxonomic and biogeographical status of guanaco *Lama guanicoe* (Artiodactyla, Camelidae). *Mammal Review*, *36*, 157–178. https://doi.org/10.1111/j.1365-2907.2006.00084.x.

González, B. A., Samaniego, H., Marín, J. C., & Estades, C. F. (2013). Unveiling current guanaco distribution in Chile based upon niche structure of phylogeographic lineages: Andean puna to subpolar forests. *PLOS One*, *8*, e78894. https://doi.org/10.1371/journal.pone.0078894.

Gordon, I. J. (2009). What is the future for wild, large herbivores in human-modified agricultural landscapes? *Wildlife Biology*, *15*, 1–9. https://doi.org/10.2981/06-087.

Hall, S. A., & Paruelo, J. M. (2006). Environmental controls on lambing rate in Patagonia (Argentina): A regional approach. *Journal of Arid Environments*, *64*, 713–735. https://doi.org/10.1016/j.jaridenv.2005.06.024.

Heard, M. J., Smith, K. F., Ripp, K. J., Berger, M., Chen, J., Dittmeier, J., Goter, M., McGarvey, S. T., & Ryan, E. (2013). The threat of disease increases as species move toward extinction: Disease and extinction. *Conservation Biology*, *27*, 1378–1388. https://doi.org/10.1111/cobi.12143.

Heuer, C., Mitchell, R. M., Schukken, Y. H., Lu, Z., Verdugo, C., & Wilson, P. R. (2012). Modelling transmission dynamics of paratuberculosis of red deer under pastoral farming conditions. *Preventive Veterinary Medicine*, *106*, 63–74. https://doi.org/10.1016/j.prevetmed.2012.02.021.

Imperiale, B., Moyano, R., Di Giulio, A., Romero, M., Alvarado Pinedo, M., Santangelo, M., Travería, G. E., Morcillo, N. S., & Romano, M. (2017). Genetic diversity of *Mycobacterium avium* complex strains isolated in Argentina by MIRU-VNTR. *Epidemiology & Infection*, *145*, 1382–1391. https://doi.org/10.1017/S0950268817000139.

Iranzo, E. C., Traba, J., Acebes, P., González, B. A., Mata, C., Estades, C. F., & Malo, J. E. (2013). Niche segregation between wild and domestic herbivores in Chilean Patagonia. *PLOS One*, *8*, e59326. https://doi.org/10.1371/journal.pone.0059326.

Jackson, J. A., Hall, A. J., Friberg, I. M., Ralli, C., Lowe, A., Zawadzka, M., Turner, A. K., Stewart, A., Birtles, R. J., Paterson, S., Bradley, J. E., & Begon, M. (2014). An immunological marker of tolerance to infection in wild rodents. *PLOS Biology*, *12*, e1001901. https://doi.org/10.1371/journal.pbio.1001901.

Judge, J., Kyriazakis, I., Greig, A., Allcroft, D. J., & Hutchings, M. R. (2020). Clustering of *Mycobacterium avium* subsp. *paratuberculosis* in rabbits and the environment: How hot is a hot spot? *Applied and Environmental Microbiology*, *71*, 6033–6038. https://doi.org/10.1128/AEM.71.10.6033-6038.2005.

Keene, O. N. (1995). The log transformation is special. *Statistics in Medicine*, *14*, 811–819. https://doi.org/10.1002/sim.4780140810.

Kennedy, D. J., & Benedictus, G. (2001). Control of *Mycobacterium avium* subsp. *paratuberculosis* infection in agricultural species. *Revue Scientifique et Technique Office International des Epizooties*, *20*, 151–179. https://doi.org/10.20506/rst.20.1.1274.

Laurin, M. L., Chaffer, M., McClure, J. T., McKenna, S. L. B., & Keefe, G. P. (2015). The association of detection method, season, and lactation stage on identification of fecal shedding in *Mycobacterium avium* ssp. *paratuberculosis* infectious dairy cows. *Journal of Dairy Science*, *98*, 211–220. https://doi.org/10.3168/jds.2014-8406.

Manning, E. J. B., & Collins, M. T. (2001). *Mycobacterium avium* subsp. *paratuberculosis*: Pathogen, pathogenesis, and diagnosis. *Revue Scientifique et Technique*, *Office International des Epizooties*, *20*, 133–150. https://doi.org/10.20506/rst.20.1.1275.

Marino, A. (2018). Dung-pile use by guanacos in eastern Patagonia. *Mammalia*, *82*, 596–599. https://doi.org/10.1515/mammalia-2017-0110.

Marsh, I., Whittington, R., & Cousins, D. (1999). PCR-restriction endonuclease analysis for identification and strain typing of *Mycobacterium avium* subsp. *paratuberculosis* and *Mycobacterium avium* subsp. *avium* based on polymorphisms in IS1311. *Molecular & Cellular Probes*, *13*, 115–126. https://doi.org/10.1006/mcpr.1999.0227.

Martin, L. B., Addison, B., Bean, A. G. D., Buchanan, K. L., Crino, O. L., Eastwood, J. R., Flies, A. S., Hamede, R., Hill, G. E., Klaassen, M., Koch, R. E., Martens, J. M., Napolitano, C., Narayan, E. J., Peacock, L., Peel, A. J., Peters, A., Raven, N., Risely, A., … Grogan, L. F. (2019). Extreme competence: Keystone hosts of infections. *Trends in Ecology & Evolution*, *34*, 303–314. https://doi.org/10.1016/j.tree.2018.12.009.

McClure, M., Machalaba, C., Zambrana-Torrelio, C., Feferholtz, Y., Lee, K. D., Daszak, P., the Future Earth oneHEALTH Global Research Project, & Karesh, W. B. (2019). Incorporating health outcomes into land-use planning. *EcoHealth*, *16*, 627–637. https://doi.org/10.1007/s10393-019-01439-x.

Mitchell, R. M., Schukken, Y., Koets, A., Weber, M., Bakker, D., Stabel, J., Whitlock, R. H., & Louzoun, Y. (2015). Differences in intermittent and continuous fecal shedding patterns between natural and experimental *Mycobacterium avium* subspecies *paratuberculosis* infections in cattle. *Veterinary Research*, *46*, 66. https://doi.org/10.1186/s13567-015-0188-x.

18651682, 2022, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/tbed.14277 by Michael Ward - University Of Sydney , Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Muskens, J., Bakker, D., de Boer, J., & van Keulen, L. (2001). Paratuberculosis in sheep: Its possible role in the epidemiology of paratuberculosis in cattle. *Veterinary Microbiology*, *78*, 101–109. https://doi.org/10.1016/S0378-1135(00)00281-9.

Navarro, D. (2015). *Learning statistics with R: A tutorial for psychology students and other beginners. version 0.5*. School of Psychology, University of Adelaide. https://cran.r-project.org/web/packages/lsr/

Naylor, A. D., Richardson, D., Sellar, M., Harley, J., Philbey, A. W., & Girling, S. J. (2018). Clinical signs, antemortem diagnostics, and pathological findings associated with *Mycobacterium avium* subspecies *paratuberculosis* infection in mishmi takin (*Budorcas taxicolor taxicolor*). *Journal of Zoo & Wildlife Medicine*, *49*, 412–419. https://doi.org/10.1638/2017-0215.1.

Nthiwa, D., Alonso, S., Odongo, D., Kenya, E., & Bett, B. (2019). Zoonotic pathogen seroprevalence in cattle in a wildlife–livestock interface, Kenya. *EcoHealth*, *16*, 712–725. https://doi.org/10.1007/s10393-019-01453-z.

Ogutu, J. O., Piepho, H.-P., Dublin, H. T., Bhola, N., & Reid, R. S. (2008). Rainfall influences on ungulate population abundance in the Mara-Serengeti ecosystem. *Journal of Animal Ecology*, *77*, 814–829. https://doi.org/10.1111/j.1365-2656.2008.01392.x.

Oliva, G., Ferrante, D., Puig, S., & Williams, M. (2012). Sustainable sheep management using continuous grazing and variable stocking rates in Patagonia: A case study. *The Rangeland Journal*, *34*, 285–295. https://doi.org/10.1071/RJ12016.

Park, H. T., Park, H. E., Park, W. B., Kim, S., Hur, T. Y., Jung, Y. H., & Yoo, H. S. (2018). Genetic diversity of bovine *Mycobacterium avium* subsp. *paratuberculosis* discriminated by IS1311 PCR-REA, MIRU-VNTR, and MLSSR genotyping. *Journal of Veterinary Science*, *19*, 627–634. https://doi.org/10.4142/jvs.2018.19.5.627

Parreño, V., Costantini, V., Cheetham, S., Blanco Viera, J., Saif, L. J., Fernández, F., Leoni, L., & Schudel, A. (2008). First isolation of rotavirus associated with neonatal diarrhoea in guanacos (*Lama guanicoe*) in the Argentinean Patagonia region: Rotavirus associated with neonatal diarrhoea in guanacos in Patagonia. *Journal of Veterinary Medicine B*, *48*, 713–720. https://doi.org/10.1111/j.1439-0450.2001.00486.x.

Pedersen, A. B., Jones, K. E., Nunn, C. L., & Altizer, S. (2007). Infectious diseases and extinction risk in wild mammals. *Conservation Biology*, *21*, 1269–1279. https://doi.org/10.1111/j.1523-1739.2007.00776.x.

Plowright, R. K., Becker, D. J., McCallum, H., & Manlove, K. R. (2019). Sampling to elucidate the dynamics of infections in reservoir hosts. *Philosophical Transactions of the Royal Society B*, *374*, 20180336. https://doi.org/10.1098/rstb.2018.0336.

Plowright, R. K., Parrish, C. R., McCallum, H., Hudson, P. J., Ko, A. I., Graham, A. L., & Lloyd-Smith, J. O. (2017). Pathways to zoonotic spillover. *Nature Reviews Microbiology*, *15*, 502–510. https://doi.org/10.1038/nrmicro.2017.45.

Pontigo, F., Radic, S., Moraga, C. A., Pulido, R., & Corti, P. (2020). Midsummer trophic overlap between guanaco and sheep in Patagonian rangelands. *Rangeland Ecology & Management*, *73*, 394–402. https://doi.org/10.1016/j.rama.2020.01.006.

R Core Team. (2021). R: A language and environment for statistical computing. R Foundation for Statistical Computing. https://www.R-project.org/

Raedeke, K. (1979). *Population dynamics and socioecology of the guanaco (Lama guanicoe) of Magallanes, Chile* (PhD dissertation). University of Washington.

Rhyan, J. C., Nol, P., Quance, C., Gertonson, A., Belfrage, J., Harris, L., Straka, K., & Robbe-Austerman, S. (2013). Transmission of brucellosis from elk to cattle and bison, Greater Yellowstone area, USA, 2002–2012. *Emerging Infectious Diseases*, *19*, 1992–1995. https://doi.org/10.3201/eid1912.130167.

Salgado, M., Herthnek, D., Bölske, G., Leiva, S., & Kruze, J. (2009). First isolation of *Mycobacterium avium* subsp. *paratuberculosis* from wild guanacos (*Lama guanicoe*) on Tierra del Fuego Island. *Journal of Wildlife Diseases*, *45*, 295–301. https://doi.org/10.7589/0090-3558-45.2.295.

Salgado, M., Aleuy, O. A., Sevilla, I., & Troncoso, E. (2015). Detection of *Mycobacterium avium* subsp. *paratuberculosis* in a cattle/pudu interface. *Arquivo Brasileiro de Medicina Veterinária e Zootecnia*, *67*, 1205–1209. https://doi.org/10.4067/S0719-81322017000200135.

Salgado, M., Manning, E. J. B., Monti, G., Bólske, G., Soderlund, R., Ruiz, M., Paredes, E., Leiva, S., Van Kruningen, H., & Kruze, J. (2011). European hares in Chile: A different lagomorph reservoir for *Mycobacterium avium* subsp. *paratuberculosis*? *Journal of Wildlife Diseases*, *47*, 734–738. https://doi.org/10.7589/0090-3558-47.3.734.

Salgado, M., Verdugo, C., Heuer, C., Castillo, P., & Zamorano, P. (2014). A novel low-cost method for *Mycobacterium avium* subsp. *paratuberculosis* DNA extraction from an automated broth culture system for real-time PCR analysis. *Journal of Veterinary Science*, *15*, 233. https://doi.org/10.4142/jvs.2014.15.2.233.

Schwarz, G. (1978). Estimating the dimension of a model. *Annals of Statistics*, *6*, 462–464. https://doi.org/10.1214/aos/1176344136.

Stevenson, K., Alvarez, J., Bakker, D., Biet, F., De Juan, L., Denham, S., Dimareli, Z., Dohmann, K., Gerlach, G. F., Heron, I., Kopecna, M., May, L., Pavlik, I., Sharp, J. M., Thibault, V. C., Willemsen, P., Zadoks, R. N., & Greig, A. (2009). Occurrence of *Mycobacterium avium* subspecies *paratuberculosis* across host species and European countries with evidence for transmission between wildlife and domestic ruminants. *BMC Microbiology*, *9*, 212. https://doi.org/10.1186/1471-2180-9-212.

Strandin, T., Babayan, S. A., & Forbes, K. M. (2018). Reviewing the effects of food provisioning on wildlife immunity. *Philosophical Transactions of the Royal Society B*, *373*, 20170088. https://doi.org/10.1098/rstb.2017.0088.

Strickland, S. J., Scott, H. M., Libal, M. C., Roussel, A. J., & Jordan, E. R. (2005). Effects of seasonal climatic conditions on the diagnosis of *Mycobacterium avium* subspecies *paratuberculosis* in dairy cattle. *Journal of Dairy Science*, *88*, 2432–2440. https://doi.org/10.3168/jds.S0022-0302(05)72921-0.

Sweeney, R. W., Collins, M. T., Koets, A. P., McGuirk, S. M., & Roussel, A. J. (2012). Paratuberculosis (Johne's disease) in cattle and other susceptible species. *Journal of Veterinary Internal Medicine*, *26*, 1239–1250. https://doi.org/10.1111/j.1939-1676.2012.01019.x.

Texeira, M., & Paruelo, J. M. (2006). Demography, population dynamics, and sustainability of the Patagonian sheep flocks. *Agricultural Systems*, *87*, 123–146. https://doi.org/10.1016/j.agsy.2004.11.005.

Thibault, V. C., Grayon, M., Boschiroli, M. L., Hubbans, C., Overduin, P., Stevenson, K., Gutierrez, M. C., Supply, P., & Biet, F. (2007). New variable-number tandem-repeat markers for typing *Mycobacterium avium* subsp. *paratuberculosis* and *M. avium* strains: Comparison with IS900 and IS1245 restriction fragment length polymorphism typing. *Journal of Clinical Microbiology*, *45*, 2404–2410. https://doi.org/10.1128/JCM.00476-07.

Turek, D., & Fletcher, D. (2012). Model-averaged Wald confidence intervals. *Computational Statistics & Data Analysis*, *56*, 2809–2815. https://doi.org/10.1016/j.csda.2012.03.002.

Verdugo, C., Pleydell, E., Price-Carter, M., Prattley, D., Collins, D., de Lisle, G., Vogue, H., Wilson, P., & Heuer, C. (2014). Molecular epidemiology of *Mycobacterium avium* subsp. *paratuberculosis* isolated from sheep, cattle and deer on New Zealand pastoral farms. *Preventive Veterinary Medicine*, *117*, 436–446. https://doi.org/10.1016/j.prevetmed.2014.09.009.

Vilá, B., & Arzamendia, Y. (2020). South American camelids: Their values and contributions to people. *Sustainability Science*. https://doi.org/10.1007/s11625-020-00874-y.

Wheeler, J.C. (1991). Origen, evolución y status actual. In S. Fernández-Baca (Ed.), *Avances y perspectivas en el conocimiento de los camélidos sudamericanos* (pp. 11-48). Santiago, Chile: FAO.

Whittington, R., & Sergeant, E. (2001). Progress towards understanding the spread, detection and control of *Mycobacterium avium* subsp *paratuberculosis* in animal populations. *Australian Veterinary Journal*, *79*, 267–278. https://doi.org/10.1111/j.1751-0813.2001.tb11980.x.

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Whittington, R. J., Marshall, D. J., Nicholls, P. J., Marsh, I. B., & Reddacliff, L. A. (2004). Survival and dormancy of *Mycobacterium avium* subsp. *paratuberculosis* in the environment. *Applied & Environmental Microbiology*, *70*, 2989–3004. https://doi.org/10.1128/AEM.70.5.2989-3004.2004.

Wilber, M. Q., Johnson, P. T. J., & Briggs, C. J. (2020). Disease hotspots or hot species? Infection dynamics in multi-host metacommunities controlled by species identity, not source location. *Ecology Letters*, *23*, 1201–1211. https://doi.org/10.1111/ele.13518.

Zhu, S., Zimmerman, D., & Deem, S. L. (2019). A review of zoonotic pathogens of dromedary camels. *EcoHealth*, *16*, 356–377. https://doi.org/10.1007/s10393-019-01413-7.

How to cite this article: Corti, P. Collado, B., Salgado, M., Moraga, C. A., Radic-Schilling, S., Tejeda, C., & Ruiz-Aravena, M. (2022). Dynamic of *Mycobacterium avium* subspecies *paratuberculosis* infection in a domestic-wildlife interface: Domestic sheep and guanaco as reservoir community. *Transboundary and Emerging Diseases*, *69*, e161–e174. https://doi.org/10.1111/tbed.14277

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